

User Experience

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Agenda

- UX Evaluation
 - General concepts
 - UX inspection + Cognitive walkthrough
 - RITE
 - UX questionnaires

UX EVALUATION GENERAL CONCEPTS

Introduction

- User Testing? No!
 - Users don't like to be tested
- Instead: UX design evaluation
 - To verify and refine designs with respect to the user experience they afford
- Also, the user involved is called a “participant” (not a subject)

Measurability of usability, user experience

- Not directly measurable
 - Like most interesting phenomena (such as teaching and learning)
- We have to measure indicators, such as time to perform a task, number of user errors
- Questionnaires provide indicators of user satisfaction, enjoyment

Types of UX evaluation data (1)

- Quantitative: Numeric data, usually from measurements (e.g., number of user errors in task performance)
 - Help evaluator assess levels of usability or UX achieved
- Qualitative: Non-numeric, descriptive data used to depict observed UX problems

Types of UX evaluation data (2)

- Objective: Data directly observed by either evaluator or a participant (e.g., measurement of time to perform a task)
- Subjective: Data based on opinion or judgment by either evaluator or participant (e.g., a user survey questionnaire)

Combinations of types of empirical UX data

- Quantitative, objective: Measures of user performance (e.g., time to do a task)
- Quantitative, subjective: Measures of opinions (e.g., questionnaire scores)
- Qualitative, objective: Description of a usability or UX problem observed
- Qualitative, subjective: Description of a usability or UX problem experienced

Formative vs. summative evaluation

- Formative evaluation helps you form design
- Summative evaluation helps you sum up design
- *“When the cook tastes the soup, that’s formative”*
- *“When the guests taste the soup, that’s summative”*

Formative evaluation

- Primarily diagnostic
- Uses qualitative data
- To identify UX problems and their causes in design
- Goal: To fix the problems

Informal summative evaluation (1)

- Partner of formative evaluation
 - Example, measurement of time on task
 - For engineering approach to summing up or assessing of UX levels
- Done without experimental controls
- CANNOT be used to make claims
 - Used for internal team use only

Informal summative evaluation (2)

- Usually without validity concerns, such as in sampling, degree of confidence
- Usually with small number of participants
 - Learn the most from each participant
- Metrics for user performance as indicators of design quality
 - Not statistically rigorous
 - Only summary statistics (e.g., mean and variance)

Analytic UX evaluation methods

- Based on looking at inherent attributes of design
- Rather than seeing design in use
- Mostly qualitative, subjective data
- Many rapid UX evaluation methods are analytic

Empirical evaluation methods

- Based on observing, experiencing, or using the design
- Employ data observed in performance of real user participants
 - Usually in UX lab or in the field

Rigor vs. speed

- Tradeoff between them
- All UX evaluation methods can span a range of rigor/rapidness
- Some methods were invented to favor rapidness over rigor

- Imagine you are a UX evaluator assigned to assess the usability of a social media platform's commenting system. This system allows users to leave comments on posts, reply to comments, and engage in discussions.
- Your goal is to identify potential UX issues. You do not have a budget to recruit participants for a user study. What do you do?

UX INSPECTIONS

COGNITIVE WALKTHROUGHS

UX inspection (1)

- Cannot inspect the user experience, but inspect design for user experience issues
- An analytical evaluation method
 - Look at design rather than usage
- Probably the primary evaluation technique in the field today

UX inspection (2)

- Variations called expert evaluation or expert inspection
- Heuristic evaluation (HE) is one specific kind of inspection (Nielsen)

UX inspection (3)

- Caution: Not always a complete substitute for user-based evaluation
- There will always be some UX problems that show up in real live user-based interaction that you will not see in any inspection

How many inspectors are needed?

- Complex question with lots of theoretical discussion
- Practical answer is “a few” (2 or 3)
 - But even one (namely you) can be effective

What kind of inspectors are needed?

- UX experts are good
- “Dual experts” are even better
 - UX expertise plus subject-matter knowledge
- Can also get this skill set by pairing up UX expert with work domain expert

Heuristic evaluation, a UX inspection method

- Drive inspection with small number (about 10) heuristics
 - Simplified, abstracted design guidelines
- Example heuristic: “Visibility of System Status”
 - The system should always keep users informed about what is going on through appropriate feedback within reasonable time.

Heuristic evaluation (1)

- Another example heuristic: “Match Between System and The Real World”
 - *The system should speak the users’ language, with words, phrases, and concepts familiar to the user rather than system-oriented terms. Follow real-world conventions, making information appear in a natural and logical order.*
- Full listing of heuristics in <https://www.nngroup.com/articles/ten-usability-heuristics/>

Heuristic evaluation (2)

- Select a small team (usually 2 or 3, could be up to 5)
- Select appropriate small set of heuristics

Heuristic evaluation (3)

- Each inspector browses through each part of UX design
 - Asking the heuristic questions
 - Assessing compliance
 - Noting where heuristics are supported and where violated, along with context (e.g., screenshots)

Heuristic evaluation (4)

- Inspectors get together as a team
- Discuss, compare, and merge problem lists
- Prioritize merged problem list
- Brainstorm suggested solutions
- Decide on recommendations
- Write report

A practical approach to UX inspection

- Definitely for UX experts, not novices
- Driven by experience
 - Not heuristics or guidelines
- But in-depth knowledge of design guidelines is essential

Practical UX inspection

- Driving perspective is usage
- Focus on
 - Tasks
 - Work activities
 - Work context
- Helps include an empirical flavor

Use a co-discovery or team approach in UX inspection

- This is one activity where two heads are almost always better than one
- Play off each other
 - Mutual sounding boards
 - Give-and-take interplay
 - Constant flow of think-aloud comments

Emotional impact inspection

- Look for fun, aesthetics, innovation
- Include packaging and out-of-the-box experience
- Try to envision long-term experience

Cognitive walkthrough

- Cognitive walkthroughs are a UX evaluation method used to assess how well users can accomplish specific tasks in a system.
- **Cognitive** – It focuses on understanding the thought processes and decision-making of users.

How does a cognitive walkthrough work

1. Select tasks/scenarios
2. Identify evaluators with user knowledge
3. Walk through the scenario
4. Evaluate the interface's user-friendliness
5. Note potential UX issues

Creating task scenarios

- Task scenarios should be realistic and representative of user goals
- Example: "Leaving a comment on a social media post."

Evaluating the User Interface

- Evaluators put themselves in the user's shoes and go step-by-step through the task scenario
- Importance of identifying issues like navigation difficulties, unclear instructions, or missing features
- Look for issues that could create an emotional response

- Are there any benefits to involving developers in the UX evaluation process?
- Think of ways to do that.

RITE

RITE: An example of a quasi-empirical method

- **Rapid Iterative Testing and Evaluation**
(Wixon et al.)
- A kind of abridged version of user-based testing
- Fast collaborative test-and-fix cycle
 - Pick low-hanging fruit
 - Relatively low cost

The RITE UX Evaluation Method (1)

- Whole team participates
- Key feature is fast turnaround
- UX problems analyzed right after evaluation with a few users

The RITE UX Evaluation Method (2)

- Whole project team decides on changes
- Changes implemented immediately
- See if the fixes worked

The RITE method: Preparation (1)

- Select UX practitioner to be facilitator
 - To direct session
- Identify characteristics needed in participants
- Decide tasks to have participants perform
 - Critical tasks, set of tasks every user must be able to perform

The RITE method: Preparation (2)

- Construct test script based on those tasks
- Decide how to collect qualitative user behavior data
- Recruit participants
 - One to three participants

The RITE method: Data collection (1)

- Gather entire project team and any other relevant project stakeholders
 - Observation room of UX lab
 - Around table in conference room
- Introduce everyone

The RITE method: Data collection (2)

- Set stage
 - Explain process
 - Explain expected outcomes
- Conduct evaluation session
 - One user participant at a time

The RITE method: Data collection (3)

- Make sure everyone knows
 - Participant is helping evaluate system
 - Team is not evaluating participant

The RITE method: Data collection (4)

- Have participant perform small number of selected tasks
 - All project stakeholders observe **silently**
 - Have participants **think aloud** as they work

The RITE method: Data collection (5)

- Work together with participants to find UX problems
 - Discover ways design should be improved
 - Take thorough notes on problem indicators
 - Example, task blocking and user errors
 - Note severity of each problem

The RITE method: Data analysis (1)

- Give list of UX problems and their causes to everyone on team
- Whole development team
 - Identify obvious problems to be fixed first
 - Example, wording or labeling are easiest fixes

The RITE method: Data analysis (2)

- Whole development team
 - Determine
 - Other problems that can reasonably be fixed
 - Problems needing more discussion
 - Problems requiring more data (from more participants)

The RITE method: Data analysis (3)

- Whole development team
 - Decide on feasible solutions for problems to be addressed
 - Implement fixes for selected problems

The RITE method: Immediate follow-up evaluation session (1)

- Bring in new participants
- Have them perform tasks associated with the fixed problems

The RITE method: Immediate follow-up evaluation session (2)

- Can determine effectiveness of changes
 - Did fix work?
 - Did fix introduce new problems?
 - Did fix uncover any hidden problems?

- You've been tasked with designing a questionnaire for assessing the UX of banking mobile application.
- Think of one question focusing on usability and one question focusing on emotional impact to add to the questionnaire.

UX QUESTIONNAIRES

Questionnaires (1)

- A self-reporting data collection technique
- Primary instrument for collecting quantitative subjective data
- Used to supplement objective data
- Or as an evaluation method on its own

Questionnaires (2)

- In past, have been used primarily to assess user satisfaction
 - But can contain probing questions about total user experience
 - Especially good for emotional impact, perceived usefulness
- Inexpensive and easy to administer
- But require skill to produce so that data are valid and reliable

Semantic differential scales (1)

- Also called Likert scales
- Each question posed on range of values describing attribute
- Most extreme value in each direction on scale is an anchor
- Scale divided with points between anchors
 - Divide up difference between anchor meanings

Semantic differential scales (2)

- Granularity of the scale
 - Number of discrete points
 - Including anchors
 - The number of choices we allow users

Semantic differential scales (3)

- Typical labeling of a point on a scale is verbal rather than just numeric
 - Often with associated numeric value
 - Labels can also be pictorial
 - Example, smiley faces
 - Helps make it language-independent

Example

- To assess participant agreement with this statement
 - *“The checkout process on this Website was easy to use.”*
- Might have these anchors: Strongly agree and Strongly disagree
- In between scale might include: Agree, Neutral, and Disagree
- Could have associated values of +2, +1, 0, -1, and -2

QUIS (1)

- Questionnaire for User Interface Satisfaction
- One of earliest and most used
- Organized around general categories: Screen, terminology, learning, system capabilities

QUIS (2)

- Many practitioners supplement QUIS with their own questions
 - Specific to design being evaluated
- There have been many extensions and variations

Example questions from QUIS

(1)

For each question, please circle the number that most appropriately reflects your impressions about this topic with respect to using this computer system or product.

1. Terminology relates to task domain:
[distantly] 0 1 2 3 4 5 6 7 8 9 10 [closely] NA
2. Instructions describing tasks [confusing] [clear]
3. Instructions are consistent [never] [always]
4. Operations relate to tasks [distantly] [closely]
5. Informative feedback [never] [always]

(2)

- 6. Display layouts simplify tasks [never] [always]
- 7. Sequence of displays [confusing] [clear]
- 8. Error messages are helpful [never] [always]
- 9. Error correction [confusing] [clear]
- 10. Learning the operation [difficult] [easy]
- 11. Human memory limitations [overwhelmed] [are respected]
- 12. Exploration of features [discouraged] [encouraged]

(3)

13. Overall reactions

[terrible] [wonderful]

[frustrating] [satisfying]

[uninteresting] [interesting]

[dull] [stimulating]

[difficult] [easy]

System Usability Scale (SUS)

- Just 10 questions
- Alternates positive and negative questions
 - Prevents answers without really considering the questions
- Five-point Likert scale

Example, SUS questions (1)

1. I think that I would like to use this system frequently
2. I found the system unnecessarily complex
3. I thought the system was easy to use
4. I would need technical support to be able to use this system
5. I found functions in this system integrated

Example, SUS questions (2)

- 6. Too much inconsistency in this system
- 7. Most people would learn to use this system very quickly
- 8. I found system very cumbersome to use
- 9. I felt very confident using the system
- 10. I needed to learn a lot of things before I could get going

System Usability Scale (SUS)

- Robust, extensively used
- Widely adapted
- In public domain
- Technology independent

The NASA-TLX

- The NASA-TLX is a subjective assessment tool used to measure the perceived mental workload experienced by users during a task or interaction
- Developed by NASA for aviation research, it has been widely adopted in various fields, including UX and human-computer interaction

Components of the NASA-TLX

- The NASA-TLX consists of six dimensions:
 - Mental Demand
 - Physical Demand
 - Temporal Demand
 - Performance
 - Effort
 - Frustration
- Users rate each dimension on a scale from 0 to 100, reflecting the perceived workload intensity

NASA Task Load Index

Hart and Staveland's NASA Task Load Index (TLX) method assesses work load on five 7-point scales. Increments of high, medium and low estimates for each point result in 21 gradations on the scales.



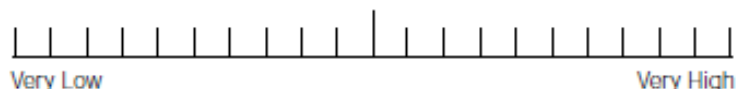
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Name	Task	Date
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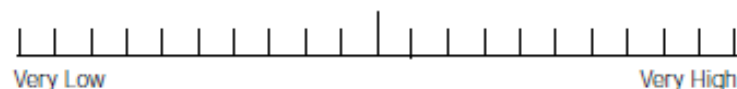
Mental Demand How mentally demanding was the task?



Physical Demand How physically demanding was the task?



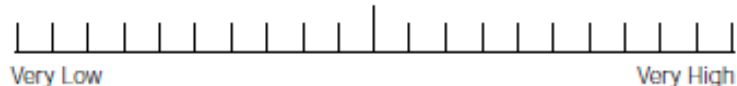
Temporal Demand How hurried or rushed was the pace of the task?



Performance How successful were you in accomplishing what you were asked to do?



Effort How hard did you have to work to accomplish your level of performance?



Frustration How insecure, discouraged, irritated, stressed, and annoyed were you?



Administering the NASA-TLX

- Data Collection: users complete the NASA-TLX questionnaire after performing a task or interaction
- Dimension Ratings: users assign ratings to each dimension based on their subjective experience
- Weighting: users assign weights to indicate the relative importance of each dimension

Interpreting NASA-TLX Results

- Calculate the weighted workload score by multiplying dimension ratings by weights and summing them
- Higher scores indicate higher perceived workload, while lower scores indicate lower workload
- The NASA-TLX results can inform design decisions, task allocation, and workload management to optimize the user experience

AttrakDiff

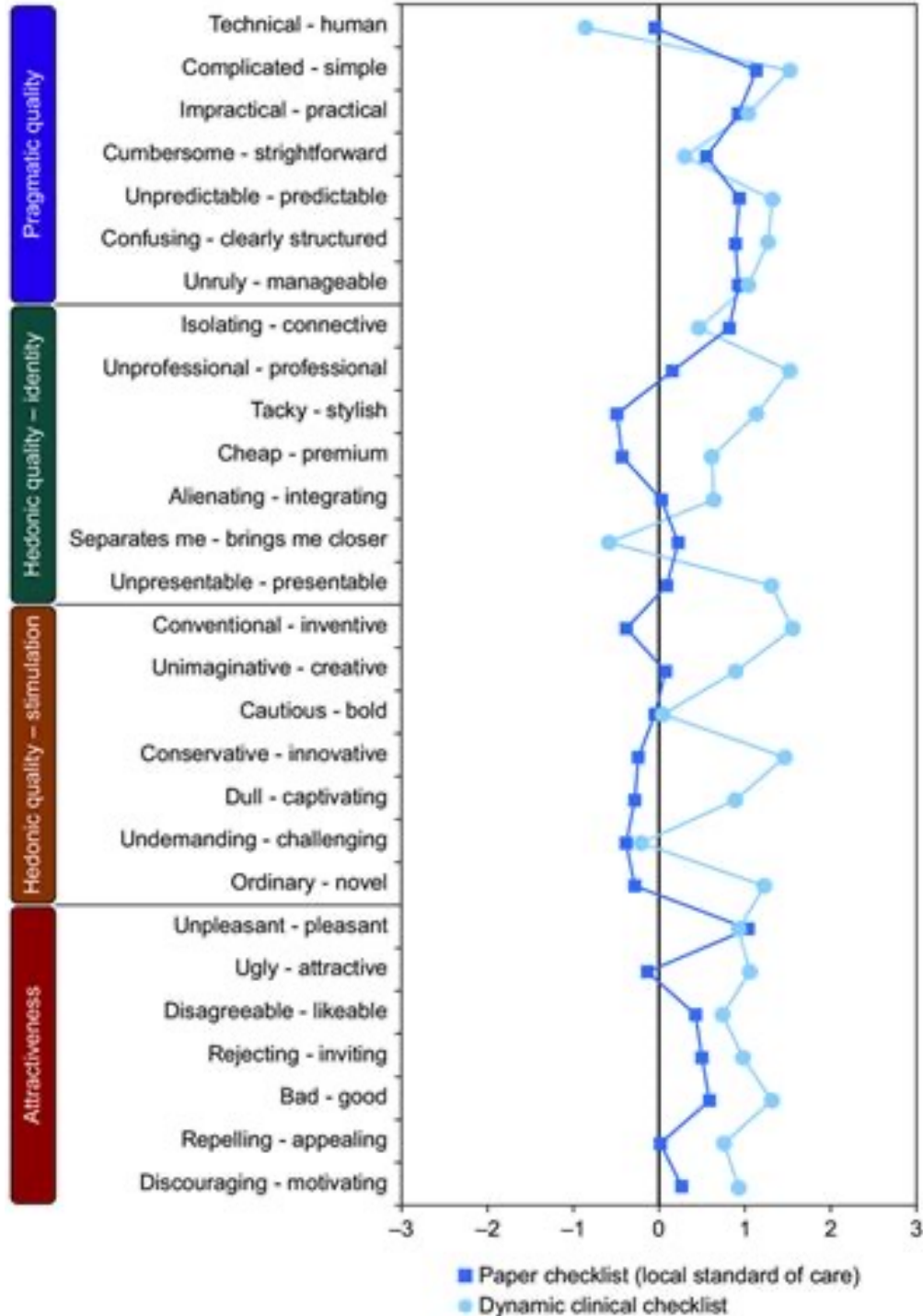
- AttrakDiff is a questionnaire used to assess the perceived quality of a user experience by measuring two dimensions: hedonic (emotional) and pragmatic (utilitarian) quality.
- Developed by Hassenzahl and colleagues, it provides insights into the overall attractiveness and usability of a product

Dimensions of AttrakDiff

- Hedonic Quality: the hedonic dimension focuses on users' emotional responses, including factors like attractiveness, novelty, and stimulation
- Pragmatic Quality: the pragmatic dimension assesses utilitarian aspects, such as efficiency, clarity, and dependability

Ratings and Scaling in AttrakDiff

- Users rate items related to hedonic and pragmatic quality on a scale from -3 (negative) to +3 (positive)
- The questionnaire separates hedonic and pragmatic dimensions for a nuanced evaluation

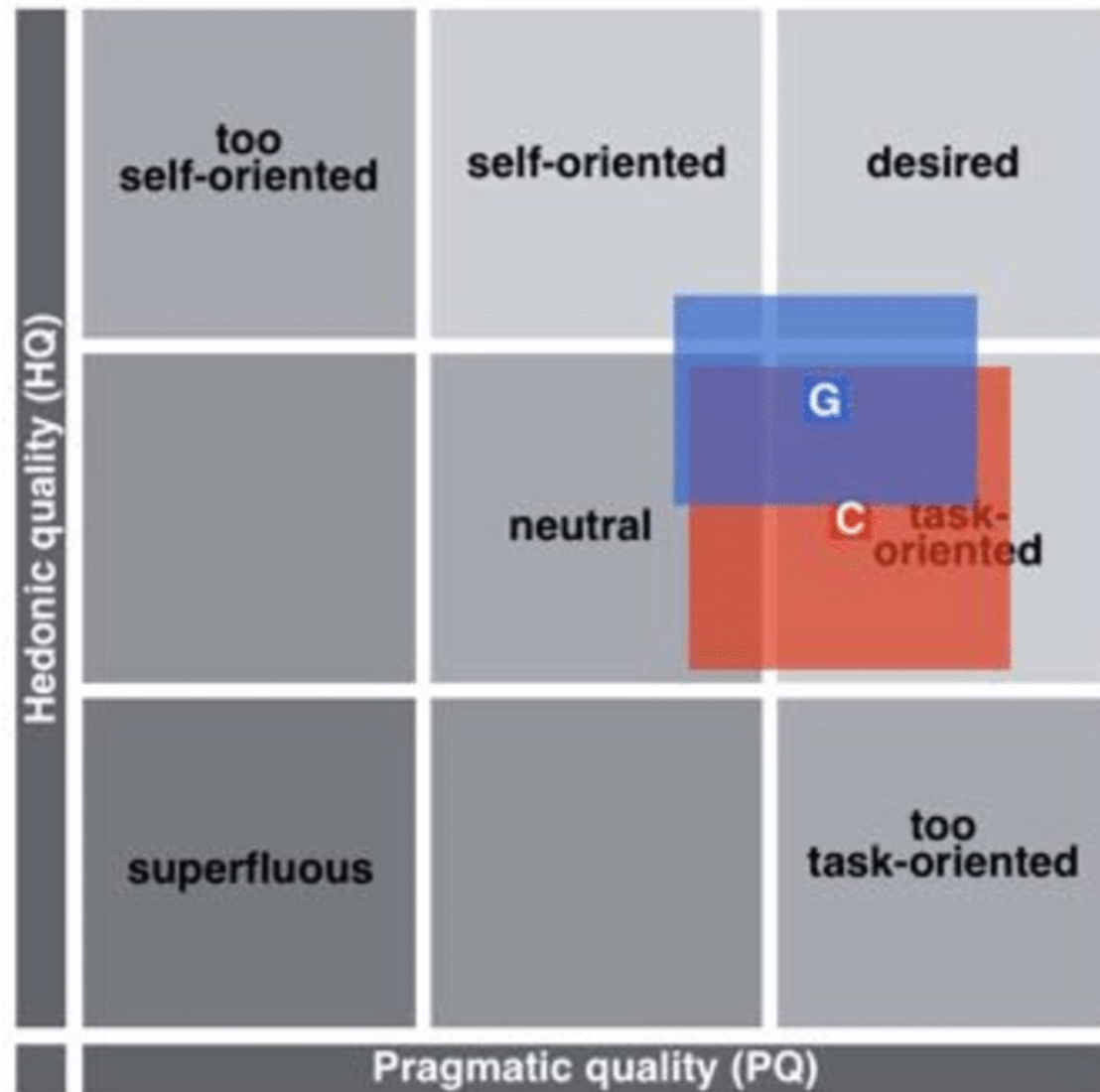


Administering AttrakDiff

- Users complete AttrakDiff after interacting with a product or system
- Scores are analyzed separately for each dimension and can provide insights into the user experience

Interpreting AttrakDiff Results

- AttrakDiff results can be visualized using a quadrant chart, with hedonic and pragmatic quality axes
- AttrakDiff results can guide design decisions by identifying areas for improvement in both emotional appeal and utilitarian usability



Adapting questionnaires

- You can modify an existing questionnaire by
 - Choosing a subset of questions
 - Changing the wording in some questions
 - Adding questions
 - Using different scale values
- Modifying a questionnaire can damage validity, but not a concern in formative evaluation

Next week

- T class
 - UX evaluation (part 2)
- TP class
 - Student presentations
 - UX evaluation